

PERSPECTIVE

Trait-based approaches and national biodiversity conservation

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Abstract

Biodiversity encompasses species diversity, functional diversity, genetic diversity, and phylogenetic diversity, which are outcomes of evolutionary and ecological processes. These biodiversity components play an important role in shaping ecosystem function and stability in distinct ways. National and international conservation programs have typically considered species diversity, while other dimensions of biodiversity are often *neglected* in conservation policy. In 2010, the Convention on Biological Diversity (CBD) established the global Strategic Plan for Biodiversity 2011–2020 and Aichi Biodiversity Targets, which aim to halt biodiversity loss, enhancing ecosystem conservation and restoration. To deliver these global strategies and targets, CBD member countries have subsequently developed and implemented national biodiversity conservation action plans. Here, we investigated the use of functional trait-based approaches in conservation policy at the national level to meet the Aichi Biodiversity Targets. Trait-based approaches promote functional diversity within communities, supporting ecosystem services and human health and well-being. We accessed the 5th and 6th National Reports submitted to the CBD over the last 11 years. We found that 26% of the countries incorporate trait-based approaches in their 5th National Reports, and 33% incorporate trait-based approaches in the 6th National Reports, highlighting widespread failure to consider functional diversity in decision making. We advocate for a greater focus on functional diversity in conservation policies for better conservation outcomes.

KEYWORDS

biodiversity conservation, conservation policy, functional diversity, functional trait, global biodiversity framework, trait-based approach

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1 | DIMENSIONS OF BIODIVERSITY AND ECOSYSTEM SERVICES

Protecting and restoring biodiversity are major goals of conservation, in part due to the intrinsic value of species, but importantly, also because biodiversity promotes ecosystem function and stability. Healthy watersheds, dynamic nutrient cycles, carbon sequestration, and primary productivity are essential for global ecosystem function and provides the ecosystem services on which humans rely, including clean air and water, food production, erosion control, and carbon storage, as well as cultural ecosystem services (Millennium Ecosystem Assessment, 2015). Often, conservation plans focus on efforts to stabilize or maximize the persistence of particular species or species numbers (Fletcher et al., 2025; Mace, 2004), with an emphasis on the population stability of rare and keystone species (Golding et al., 2024; Marcot & Flather, 2007; Russo & Dechmann, 2025). Such species-based approaches are well-established and immensely valuable in biodiversity conservation planning and policy decision-making. However, biodiversity encompasses three dimensions at the scale of communities: taxonomic diversity (hereafter, species diversity), phylogenetic diversity, and functional diversity, and each of these dimensions shapes ecosystem function, stability, and services, often in distinct ways (Jarzyna & Jetz, 2016; Le Bagousse-Pinguet et al., 2019). Biodiversity can also be measured at smaller (i.e., within species, via genetic diversity, Hoban et al., 2022) and larger (i.e., ecosystem diversity) scales, but those are beyond the scope of this paper.

Species diversity is the simplest dimension to measure, with species richness (the total number of species in a community) as the most commonly used biodiversity metric in conservation planning. (Hoban et al., 2022). Phylogenetic diversity involves quantifying the relatedness among species in a community using molecular data, and this dimension reflects evolutionary processes in deep time (Gerhold et al., 2018). Phylogenetic and genetic diversity are less commonly incorporated into conservation planning, despite important implications for community stability (Robuchon et al., 2023). Functional diversity is estimated from species traits relating to resource use, species interactions, and stress tolerance, and it is sometimes inferred from species richness, although this relies on the assumption that species occupy distinct niches such that species number correlates with functional diversity (Cadotte et al., 2011). Increasingly, it is becoming clear that species and functional diversity are often not correlated as a consequence of community assembly filters, variable disturbance regimes, dynamic species interactions, and environmental heterogeneity associated with spatial

scale (Suárez-Castro et al., 2022). Considering that species and functional diversity are often uncorrelated, they provide complementary insights into the role of biodiversity in ecosystem function and services (Greenop et al., 2021), and for that reason, the incorporation of functional diversity into conservation planning and implementation may improve conservation outcomes.

2 | FUNCTIONAL TRAIT-BASED APPROACHES FOR CONSERVATION

Higher species richness is predicted to contribute to ecosystem function and stability because of complementary roles of species but also because of redundancy, which allows communities to be resilient even in the face of some biodiversity loss (Wagg et al., 2022). To better understand the mechanisms of complementarity and redundancy, characterizing species' functional strategies is informative, particularly in light of global environmental change (Díaz et al., 2007). Species can be characterized on the basis of functional traits, measurable features of the anatomy, morphology, physiology, architecture, and phenology, which contribute to plant performance and their ecosystem effects (Violle et al., 2007). Variation in functional traits within and among species within communities defines the functional diversity that shapes ecosystem function and services (Díaz & Cabido, 2001). Growing evidence suggests that functional diversity contributes more than species richness to some ecosystem services, such as regulating water flow, nutrient cycling, reducing soil erosion, increasing carbon sequestration, enhancing agricultural productivity, and providing cultural ecosystem services, illustrating the importance of functional biodiversity for human livelihood (Graves et al., 2017; Satdichanh et al., 2023; Zhu et al., 2015).

Trait-based approaches are increasingly used in conservation practices, particularly within restoration activities (Gornish et al., 2023). Plant functional traits provide mechanistic insights into how plants tolerate environmental stressors, interact with other organisms, and drive nutrient and water cycling in the ecosystem (Violle et al., 2007). A mechanistic perspective in conservation planning via trait-based approaches is attractive because traits allow for a predictive framework. For example, if we know which traits contribute to climate stress tolerance (O'Brien et al., 2017), or carbon sequestration (Miedema Brown & Anand, 2022), or food provisioning (Cámara-Leret et al., 2017), we can design conservation projects with species that have these traits to meet conservation goals. In addition, we can use traits to monitor plant performance during periods of stress in order to identify thresholds in decline at which intervention is

necessary (Schönbeck et al., 2023). These are the most common ways that trait-based approaches are used in conservation: to inform species selection to meet restoration goals, and to monitor the performance of focal species (Merchant et al., 2023).

National and international biodiversity monitoring programs commonly measure species diversity and composition, but tend to overlook trait-based functional diversity. Failure to consider functional diversity in biodiversity conservation can lead to its unintentional loss with a corresponding decline in ecosystem services (Flynn et al., 2009; Le Provost et al., 2020). To improve conservation outcomes, we advocate for a greater focus on functional diversity to inform species selection, develop monitoring programs, support Indigenous stewardship, and as a formal target in conservation policies.

3 | TRAIT-BASED APPROACHES AND NATIONAL CONSERVATION POLICY

In 2010, during the 10th Conference of the Parties to the United Nation's Convention on Biological Diversity (CBD), many countries adopted The Strategic Plan for Biodiversity 2011–2020 and a set of 20 Aichi Biodiversity Targets (Aichi Targets). This 10-year framework aims to halt the loss of biodiversity and guide global biodiversity conservation between 2011 and 2020 (Convention of Biological Diversity, 2010). Each country member of the CBD is required to develop a National Biodiversity Strategy and Action Plan (NBSAP) that addresses threats to biodiversity, specific to their local, regional, and national needs. Importantly, NBSAPs can be used as a key policy tool to coordinate biodiversity actions across public and private sectors by embedding biodiversity conservation and sustainable use into national policies, strategies, and conservation practices. NBSAPs were meant to act as the *main vehicle* to ensure implementation of plans to reach the goals of the Strategic Plan and Aichi Targets, increasing awareness and strengthening biodiversity conservation across many regions (Cardona Santos et al., 2023). Members of the CBD are required to submit a National Report to provide information on NBSAP implementation, to inform the status of biodiversity, and to identify research and policy gaps in biodiversity conservation at the national level. The 5th National Report provides the mid-term evaluation of the NBSAP implementation, while the 6th National Report provides the final evaluation of the NBSAP implementation and progress toward the Strategic Plan for Biodiversity 2011–2020 and Aichi Targets (Buchanan et al., 2020). Despite some progress, none of the Aichi Targets were fully achieved, and only a

few targets were partially met, indicating the need to enhance NBSAP planning and implementation with clearer and measurable Targets and better scientific and technical cooperation (Secretariat of the Convention on Biological Diversity, 2020).

Biodiversity conservation in the era of global change is challenging, due to rapid changes in land-use, environmental conditions, and climate. Among the highest conservation priorities are global biodiversity hotspots because these have unique biotas with high rates of endemism (Myers et al., 2000). Biodiversity hotspots also have characteristically narrow geographic extents, leading to the proposed biodiversity hotspot concept for an area-based approach to preserve and restore the *most species at the least cost* (Myers et al., 2000). The Strategic Plan for Biodiversity 2011–2020 and Aichi Targets prioritize ecosystem-level conservation and restoration (Targets 5, 6, 7, 14, and 15), to prevent the spread of invasive species (Target 9), to protect threatened species (Target 12), and to share biodiversity data (its status, value and function, Target 19). Although trait-based approaches are not explicitly identified within the Aichi framework and targets, they can be used to support its planning and implementation. Currently, all member countries of CBD are transitioning their national biodiversity strategies and action plans from Aichi Biodiversity Targets to the most recent Global Biodiversity Framework (CBD, 2022). Our aim here is to investigate the use of trait-based approaches in the previous conservation policy at the national level, in order to provide a baseline and identify opportunities to incorporate trait-based approaches in future conservation planning and implementation.

To determine the extent to which functional traits are already used in national conservation planning and implementation documents, we assessed the 5th and 6th National Reports submitted to the CBD between 2014 and 2020, focusing specifically on 43 countries (Table S1) located within biodiversity hotspots (Myers et al., 2000). The selected countries include nations of varying income levels, in order to capture the full range of financial capacity for support of biodiversity conservation activities. Please note that, at the time this assessment was conducted, the 7th National Report regarding the implementation of the current Global Biodiversity Framework had not yet been submitted to the CBD. We surveyed the National Reports using search terms for “trait” applying a binary approach to identify any use of trait-based approaches (for species selection, monitoring, or functional diversity metrics). Although not all trait-based restoration and conservation activities explicitly use the term “trait,” our aim here was to identify formalized trait-based approaches as part of national conservation

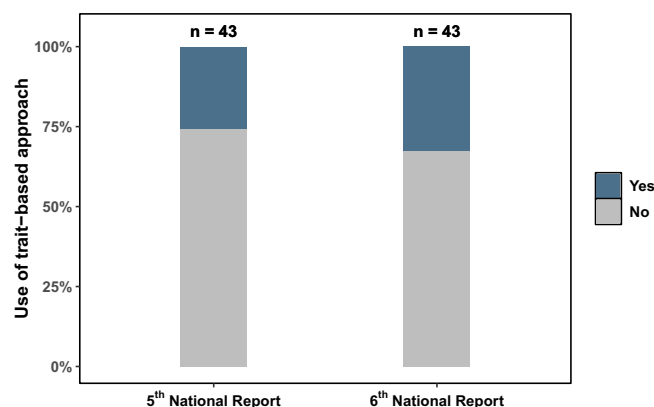


FIGURE 1 The use of trait-based approach in biodiversity conservation at the national level. 5th and 6th National Report are focusing on 43 countries located within the biodiversity hotspots for conservation priorities.

TABLE 1 Example of open access trait databases that can be used for national biodiversity conservation planning and implementation.

	Open access database	References
Plant	TRY plant trait database; BIEN Botanical Information and Ecological Network; Aus Trait	Kattge et al. (2020); Enquist et al. (2016); Falster et al. (2021)
Terrestrial animal	AnimalTraits; An amniote life-history database; and AVONET	Herberstein et al. (2022), Myhrvold et al. (2015), Tobias et al. (2022)
Fresh water fish	FISHMORPH	Brosse et al. (2021)
Coral species	Coral Trait Database	Madin et al. (2017)
Marine species	Marine Species Traits	Marine Species Traits Editorial Board (2026)
Other open access trait-data repositories	Open Traits Network	

planning and implementation. We noted that alternative terms for “trait” may have been used by practitioners and decision-makers such as “characteristic” or “feature.” However, these other terms do not necessarily indicate the use of a trait-based approach, and so detection of these terms would lead to an over-estimation of the use of trait-based approaches (via false positives, leading to a high risk of Type-I error in our results), and therefore we

excluded other terms from our searches. We found that only 26% of the countries incorporated traits in their 5th National Reports, while 33% did so in the 6th National Reports, revealing a slight increase in trait-based approaches (Figure 1). A surprising 74% (5th National Reports) and 67% (6th National Reports) of the countries do not mention species traits at all in their Action Plans. The infrequent use of trait-based approaches to meet conservation goals highlights an opportunity for greater use of functional traits to inform conservation.

Barriers to the use of trait-based approaches in conservation plans include the specialized instruments and knowledge required for trait analysis. However, trait data feature prominently in the ecological literature, and associated with the increase in trait-based research, there has been rapid growth of archived trait data for plants, as well as animals (Table 1). Practitioners and policymakers have unlimited access to these trait databases, providing a vital step toward integrating trait-based approaches into biodiversity conservation planning at national levels worldwide. Nonetheless, for some applications, there may be limitations in trait data availability due to research biases. Woody plants in temperate ecosystems are commonly measured (Kattge et al., 2020), which may limit trait data coverage within other ecosystems (Bridge et al., 2020). Moreover, traits are highly variable within species (Westerband et al., 2021), and care should be taken when extrapolating from the site at which traits were sampled to inform conservation in a different place. Trait data should be used intentionally to represent the species and sites of interest (Májeková et al., 2016).

4 | GLOBAL EFFORTS TO ORGANIZE BIODIVERSITY CONSERVATION PLANS

Human livelihood is critically tied to biodiversity and ecosystem function. The loss of biodiversity can cause a significant impact on human health and well-being (Cardinale et al., 2012). Globally, approximately 28% of biodiversity (IUCN, 2025), or more than one million species, are threatened and facing extinction risk (IPBES, 2019). This marks the most severe biodiversity crisis in human history. Changes in land- and sea-use, over-exploitation, climate change, and invasive species are reported as key direct drivers of the current anthropogenic biodiversity loss (Jaureguiberry et al., 2022).

Following the failure of Strategic Plans for Biodiversity 2011–2020 and Aichi Targets, members of the CBD have launched a renewed conservation effort to address the biodiversity crisis. This new global conservation program aims to provide clearer conservation goals and targets and

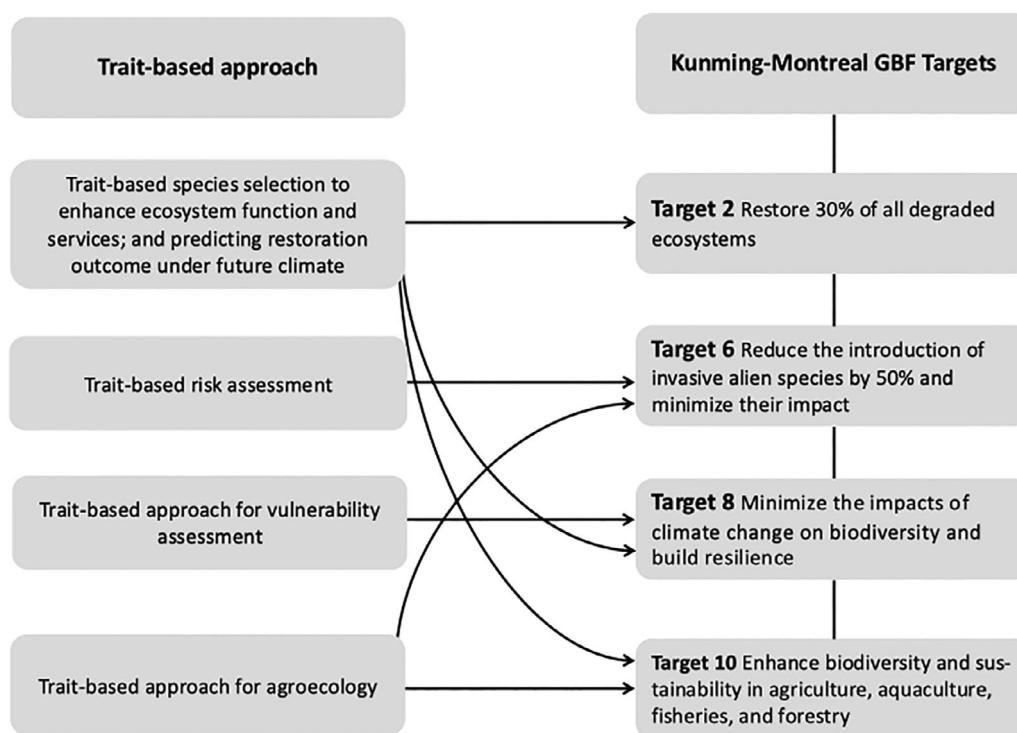


FIGURE 2 Conceptual ideas, integrating trait-based approaches in implementing Kunming-Montreal Global Biodiversity Framework at the national level.

to select a framework that covers all aspects of the United Nations Sustainable Development Goals (Obura, 2023). The Kunming-Montreal Global Biodiversity Framework (GBF) was adopted by 196 countries in December 2022 (CBD, 2022), during the 15th Conference of the Parties to the United Nations Convention on Biological Diversity (CBD). To achieve CBD's 2050 vision to "live in harmony with nature," the Kunming-Montreal framework is comprised of four main goals: (1) protect and restore, (2) prosper with nature, (3) share benefits fairly, and (4) invest and collaborate. To meet these goals, there are 23 implementation targets organized into three groups with the aims to: reduce the threats to biodiversity (targets 1–8), meet people's needs via sustainable use of biodiversity and fair benefits sharing (targets 9–13), and harmonize across public and private sectors by mobilizing available resources and filling the finance gaps (targets 14–23). Targets 2 and 3 are the most widely discussed and celebrated of all targets, stating that by 2030, at least 30% of the global degraded ecosystems be "under effective restoration," and that at least 30% of the terrestrial, inland water, coastal, and marine ecosystems be protected and managed in ways that "recognize and respect the rights of Indigenous peoples and local communities." The Kunming-Montreal GBF marks one of the most ambitious global biodiversity plans since the Paris Agreement and UN biodiversity strategic plan for 2011–2020 attempted to reverse biodiversity loss and mitigate the

impacts of climate change. The GBF is a promising effort to enhance global biodiversity conservation, but without clear guidance and assessment plans, there is concern that the Kunming-Montreal GBF may just be another business venture rather than an authentic turning point to halt biodiversity losses (Obura, 2023).

Similar to the implementation of Aichi Targets, each country member of CBD is required to develop a National Biodiversity Strategy and Action Plan (NBSAP) that is aligned with the Kunming-Montreal framework and targets. NBSAP acts as a key policy tool in implementing the new global biodiversity framework, specifically integrating biodiversity conservation across all sectors, while maintaining the sustainable interactions between human activities and biodiversity. Despite the growing recognition of trait-based approaches as important tools in conservation and restoration, their application in conservation policies remains limited (Miatta et al., 2021). Many countries have not incorporated trait-based approaches into national conservation planning and implementation to achieve the Aichi Targets (Figure 1). This represents a missed opportunity to formally address the role of functional diversity in ecosystem function and could lead to unintended declines in ecosystem services. There are various opportunities to integrate trait-based approaches into Kunming-Montreal targets (Figure 2). Target 2 focuses on ecosystem restoration, and traits are routinely used to select species with

target functions that contribute to ecosystem services and to monitor outplant performance (Merchant et al., 2023). Target 6 aims to reduce invasive species spread, which can be informed by trait-based risk assessment tools that identify potential invaders and predict future invasion success (Chan et al., 2021). Target 8 aims to enhance climate change resilience in biological communities and ecosystems. Functional traits feature prominently in restoration and conservation plans to address climate change (Barton et al., 2026; Gallagher et al., 2021). Species with known climate stress tolerance can be leveraged for restoration, while sensitive species can be identified for monitoring and management (Harper et al., 2022). Target 10 focuses on provisioning ecosystem services, and the integration of trait-based approaches can be used to inform farmers' decisions, boost productivity, and enhance provisioning ecosystem services in managed landscapes (Damour et al., 2018). The main goal of the United Nations biodiversity initiatives is to reverse biodiversity loss by linking sustainable development and biodiversity conservation. Another key concept of Kunming-Montreal framework is to recognize the rights of Indigenous communities and to feature Indigenous knowledge in biodiversity conservation globally (Danielsen et al., 2024). We propose that functional trait approaches can complement Indigenous methods, which are known to be highly effective for biodiversity conservation (Niu et al., 2022).

5 | CONCLUSION

Despite growing evidence highlighting the important links between functional traits, ecosystem services, and human well-being, the application of trait-based approaches in conservation and policy decision-making has yet to be fully realized or embraced by policy-makers. The consequence is the development of conservation strategies and frameworks that narrowly focus on the preservation of individual species, rather than addressing the species' functional roles within ecosystems. The latest global biodiversity initiatives (Kunming-Montreal Framework) aim to reduce biodiversity loss, maintain the needs of local and Indigenous people while protecting and restoring ecosystems. As member countries of CBD update and submit their (new) National Biodiversity Strategies and Action Plans (NBSAPs) that align with the goals and targets of Kunming-Montreal Global Biodiversity Framework, they have a great opportunity to finalize the NBSAPs and incorporate scientific approaches into national conservation policy. We call for action that trait-based approaches should be incorporated into NBSAPs planning and implementation to promote the role of functional diversity in ecosystem function and services, to meet Kunming-Montreal goals and targets.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data supported the finding of this paper can be found in the Supporting Information.

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SUPPORTING INFORMATION

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